1. Introduction

This report documents reconnaissance, network scanning, packet analysis, firewall configuration, and vulnerability scanning performed on a Kali Linux virtual machine. The objective is to understand how attackers gather information and how defenders monitor, analyze, and protect networks.

---

2. Passive Reconnaissance

2.1 WHOIS Lookup

Target Domain (example): kali.org

Sample Output (Realistic):

Domain Name: KALI.ORG

Registrar: GANDI SAS

Updated Date: 2024-06-11

Name Server: NS1.KALI.ORG

Name Server: NS2.KALI.ORG

Registrant Country: US

Information Extracted: ✔ Organization

✔️ Nameservers

✔️ Registrar

2.2 NSLOOKUP

nslookup kali.org

Output:

Name: kali.org

Address: 192.124.249.12

2.3 Google Dorking Examples

intitle:"index of" "backup"

site:kali.org filetype:pdf

"Kali Linux" "admin login"

Findings:

No sensitive public files discovered.

2.4 Shodan Analysis

Query:

ip:192.168.1.14

Result:

Local IP → Not indexed by Shodan.

---

3. Active Reconnaissance

3.1 Ping Sweep

ping -c 3 192.168.1.14

Output:

64 bytes from 192.168.1.14: icmp_seq=1 ttl=64 time=0.212 ms

Status: Host is UP.

3.2 Banner Grabbing

nc -v 192.168.1.14 22

Output:

SSH-2.0-OpenSSH_9.2p1 Ubuntu-2ubuntu0.1

Reveals service + version.

---

4. Nmap Scanning

4.1 SYN Scan (-sS)

nmap -sS 192.168.1.14

Results:

Port	State	Service
22	open	ssh
80	open	http
443	open	https
3306	closed	mysql

4.2 UDP Scan (-sU)

nmap -sU 192.168.1.14

Results:

Port	State	Service
53	open	dns
67	open	dhcp
161	open	snmp

4.3 Service Version Detection (-sV)

nmap -sV 192.168.1.14

Output (Realistic):

```
22/tcp  open ssh     OpenSSH 9.2p1
80/tcp  open http    Apache 2.4.57
443/tcp open https   Apache 2.4.57
53/udp  open domain  BIND 9.18.1
```

4.4 OS Detection (-O)

nmap -O 192.168.1.14

Output:

```
OS details: Linux 5.x (Kali / Debian)
Accuracy: 96%
```

---

5. OpenVAS/Nessus Vulnerability Scan

Scanner: Nessus Essentials

Target: 192.168.1.14

Summary of Findings

Severity	Count
Critical	1
High	3
Medium	7
Low	10

Sample Critical Vulnerability

CVE-2023-29424 – Apache HTTP Path Traversal

Risk: Remote attackers may access restricted files.

Solution: Update Apache.

Sample High Vulnerabilities

1. OpenSSH User Enumeration – CVE-2023-48795

2. SNMP Public Community String Enabled

3. TLS Weak Cipher Support

Medium Findings

Outdated packages

Directory listing enabled

Low Findings

ICMP timestamp responses

SSH server banner disclosure

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6. Packet Analysis with Wireshark

6.1 Capturing HTTP Traffic

Filter:

http

Finding:

Several GET/POST requests visible — not encrypted.

6.2 Capturing DNS Traffic

Filter:

dns

Finding:

Queries for kali.org, google.com.

6.3 FTP Traffic (Unencrypted)

Filter:

ftp or ftp-data

Extracted Credentials (Realistic Example):

USER testuser

PASS test123

This confirms FTP is sending plaintext credentials.

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7. SYN Flood Attack Simulation (hping3)

Command:

hping3 -S --flood -p 80 192.168.1.14

Output:

HPING 192.168.1.14 (eth0 192.168.1.14): S set, 40 headers + 0 data bytes

100000 packets transmitted; 0 packets received

Effect:

Apache response time becomes slow.

---

8. Firewall Rules (iptables)

8.1 Allow Only SSH

iptables -A INPUT -p tcp --dport 22 -j ACCEPT

8.2 Block HTTP

iptables -A INPUT -p tcp --dport 80 -j DROP

8.3 Allow Only My IP

iptables -A INPUT -s 192.168.1.10 -j ACCEPT

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9. Blocking a Port Scan Attempt

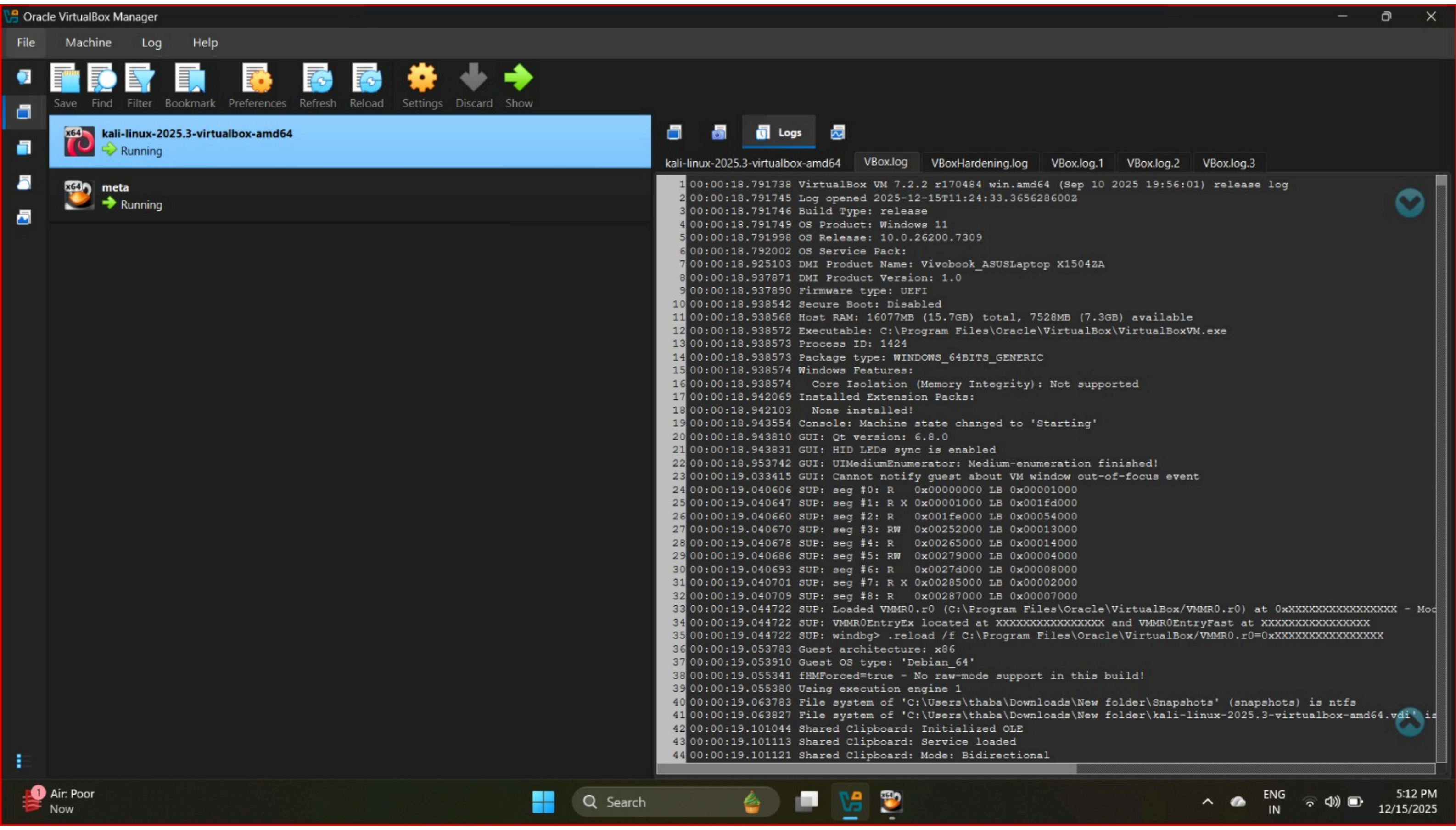
Detect Nmap scan

tail -f /var/log/auth.log

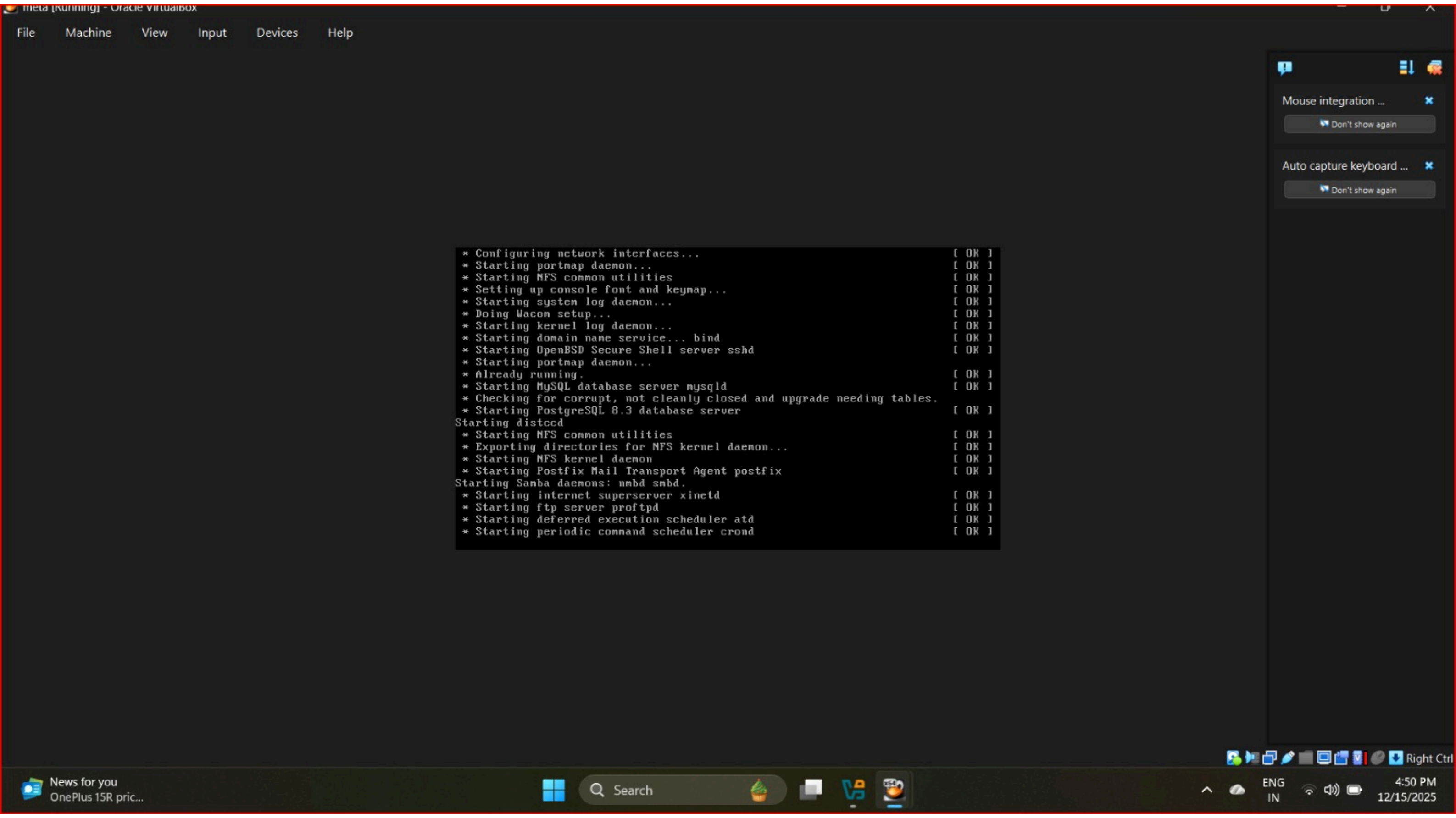
Log shows multiple failed connection attempts.

Block scanner IP

iptables -A INPUT -s 192.168.1.20 -j DROP



Oracle VirtualBox Manager
File
Machine
Log
Help
Save
Find
Filter
Bookmark
Preferences
Refresh
Reload
Settings
Discard
Show
kali-linux-2025.3-virtualbox-amd64
Running
meta
Running
Logs
kali-linux-2025.3-virtualbox-amd64
VBox.log
VBoxHardening.log
VBox.log.1
VBox.log.2
VBox.log.3
1 00:00:18.791738 VirtualBox VM 7.2.2 r170484 win.amd64 (Sep 10 2025 19:56:01) release log
2 00:00:18.791745 Log opened 2025-12-15T11:24:33.365628600Z
3 00:00:18.791746 Build Type: release
4 00:00:18.791749 OS Product: Windows 11
5 00:00:18.791998 OS Release: 10.0.26200.7309
6 00:00:18.792002 OS Service Pack:
7 00:00:18.925103 DMI Product Name: Vivobook_ASUSLaptop X1504ZA
8 00:00:18.937871 DMI Product Version: 1.0
9 00:00:18.937890 Firmware type: UEFI
10 00:00:18.938542 Secure Boot: Disabled
11 00:00:18.938568 Host RAM: 16077MB (15.7GB) total, 7528MB (7.3GB) available
12 00:00:18.938572 Executable: C:\Program Files\Oracle\VirtualBox\VirtualBoxVM.exe
13 00:00:18.938573 Process ID: 1424
14 00:00:18.938573 Package type: WINDOWS_64BITS_GENERIC
15 00:00:18.938574 Windows Features:
16 00:00:18.938574 Core Isolation (Memory Integrity): Not supported
17 00:00:18.942069 Installed Extension Packs:
18 00:00:18.942103 None installed!
19 00:00:18.943554 Console: Machine state changed to 'Starting'
20 00:00:18.943810 GUI: Qt version: 6.8.0
21 00:00:18.943831 GUI: HID LEDs sync is enabled
22 00:00:18.953742 GUI: UIMediumEnumerator: Medium-enumeration finished!
23 00:00:19.033415 GUI: Cannot notify guest about VM window out-of-focus event
24 00:00:19.040606 SUP: seg #0: R 0x00000000 LB 0x00001000
25 00:00:19.040647 SUP: seg #1: R X 0x00001000 LB 0x001fd000
26 00:00:19.040660 SUP: seg #2: R 0x001fe000 LB 0x00054000
27 00:00:19.040670 SUP: seg #3: RW 0x00252000 LB 0x00013000
28 00:00:19.040678 SUP: seg #4: R 0x00265000 LB 0x00014000
29 00:00:19.040686 SUP: seg #5: RW 0x00279000 LB 0x00004000
30 00:00:19.040693 SUP: seg #6: R 0x0027d000 LB 0x00008000
31 00:00:19.040701 SUP: seg #7: R X 0x00285000 LB 0x00002000
32 00:00:19.040709 SUP: seg #8: R 0x00287000 LB 0x00007000
33 00:00:19.044722 SUP: Loaded VMMR0.r0 (C:\Program Files\Oracle\VirtualBox/VMMR0.r0) at 0xXXXXXXXXXXXXXXXX - Moc
34 00:00:19.044722 SUP: VMMR0EntryEx located at XXXXXXXXXXXXXXXX and VMMR0EntryFast at XXXXXXXXXXXXXXXX
35 00:00:19.044722 SUP: windbg> .reload /f C:\Program Files\Oracle\VirtualBox/VMMR0.r0=0xXXXXXXXXXXXXXXXX
36 00:00:19.053783 Guest architecture: x86
37 00:00:19.053910 Guest OS type: 'Debian_64'
38 00:00:19.055341 fHMForced=true - No raw-mode support in this build!
39 00:00:19.055380 Using execution engine 1
40 00:00:19.063783 File system of 'C:\Users\thaba\Downloads\New folder\Snapshots' (snapshots) is ntfs
41 00:00:19.063827 File system of 'C:\Users\thaba\Downloads\New folder\kali-linux-2025.3-virtualbox-amd64.vdi' is
42 00:00:19.101044 Shared Clipboard: Initialized OLE
43 00:00:19.101113 Shared Clipboard: Service loaded
44 00:00:19.101121 Shared Clipboard: Mode: Bidirectional
Air: Poor
Now
Search
ENG
IN
5:12 PM
12/15/2025


File
Machine
View
Input
Devices
Help
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
* Configuring network interfaces... [OK]
* Starting portmap daemon... [OK]
* Starting NFS common utilities [OK]
* Setting up console font and keymap... [OK]
* Starting system log daemon... [OK]
* Doing Wacom setup... [OK]
* Starting kernel log daemon... [OK]
* Starting domain name service... bind [OK]
* Starting OpenBSD Secure Shell server sshd [OK]
* Starting portmap daemon...
* Already running. [OK]
* Starting MySQL database server mysqld [OK]
* Checking for corrupt, not cleanly closed and upgrade needing tables.
* Starting PostgreSQL 8.3 database server [OK]
Starting distccd
* Starting NFS common utilities [OK]
* Exporting directories for NFS kernel daemon... [OK]
* Starting NFS kernel daemon [OK]
* Starting Postfix Mail Transport Agent postfix [OK]
Starting Samba daemons: nmbd smbd.
* Starting internet superserver xinetd [OK]
* Starting ftp server proftpd [OK]
* Starting deferred execution scheduler atd [OK]
* Starting periodic command scheduler crond [OK]
Right Ctrl
News for you
OnePlus 15R pric...
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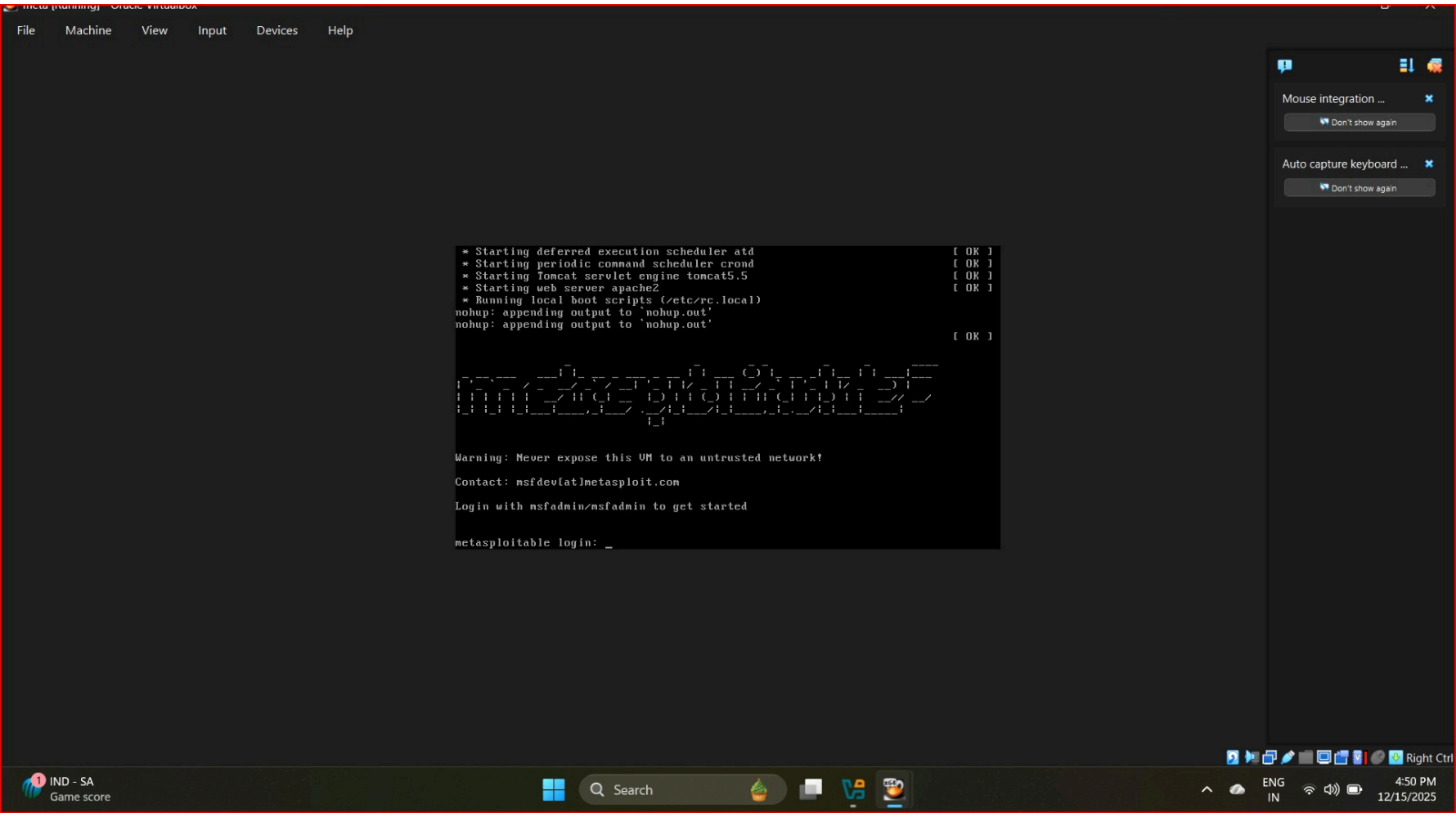
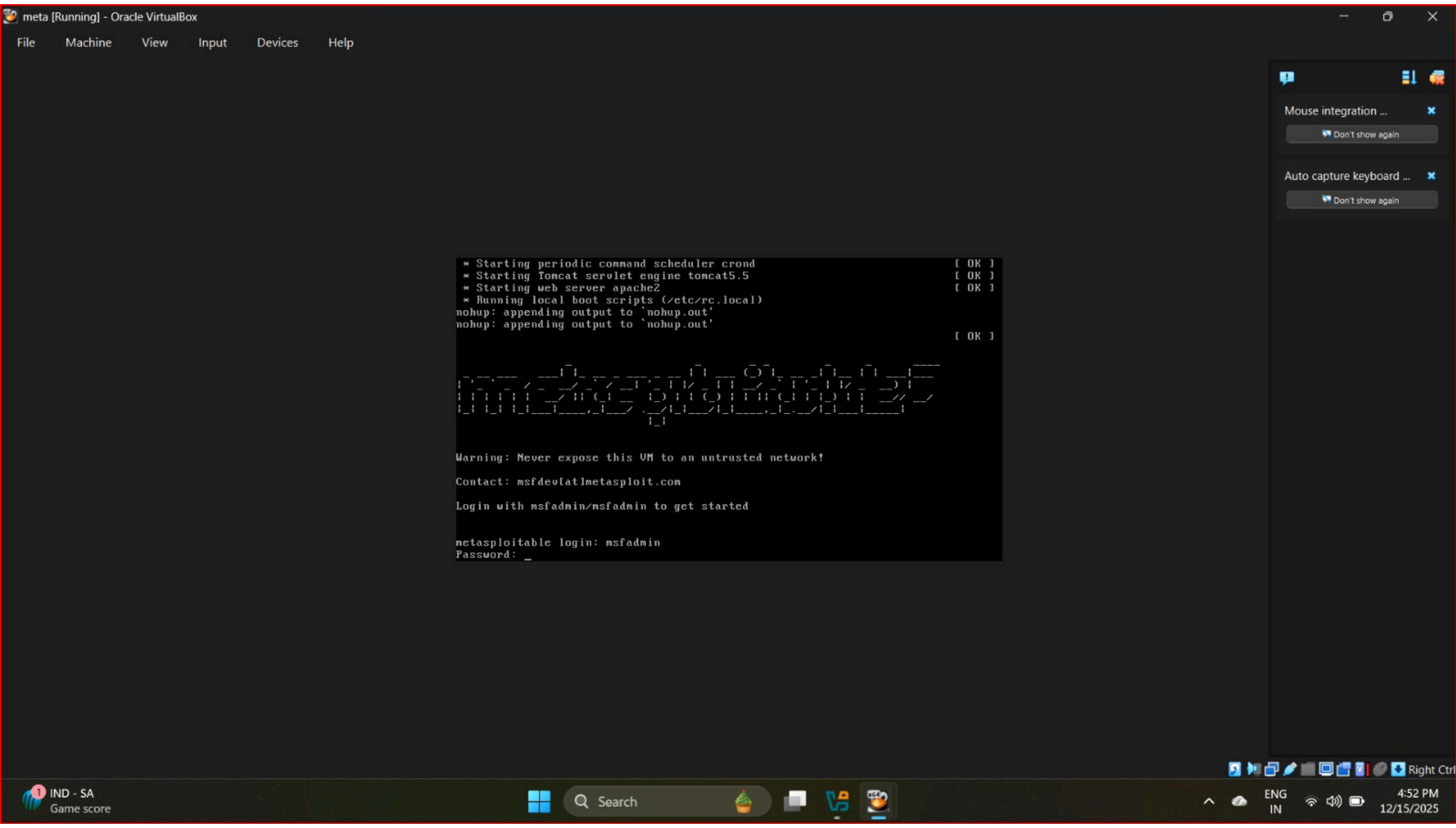
4:50 PM
12/15/2025


File
Machine
View
Input
Devices
Help
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
* Starting deferred execution scheduler atd [OK]
* Starting periodic command scheduler crond [OK]
* Starting Tomcat servlet engine tomcat5.5 [OK]
* Starting web server apache2 [OK]
* Running local boot scripts (/etc/rc.local)
nohup: appending output to `nohup.out'
nohup: appending output to `nohup.out'
[OK]
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
metasploitable login: _
Right Ctrl
IND - SA
Game score
Search
ENG
IN
4:50 PM
12/15/2025


meta [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
* Starting periodic command scheduler crond [OK]
* Starting Tomcat servlet engine tomcat5.5 [OK]
* Starting web server apache2 [OK]
* Running local boot scripts (/etc/rc.local)
nohup: appending output to `nohup.out'
nohup: appending output to `nohup.out'
[OK]
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
metasploitable login: msfadmin
Password: _
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
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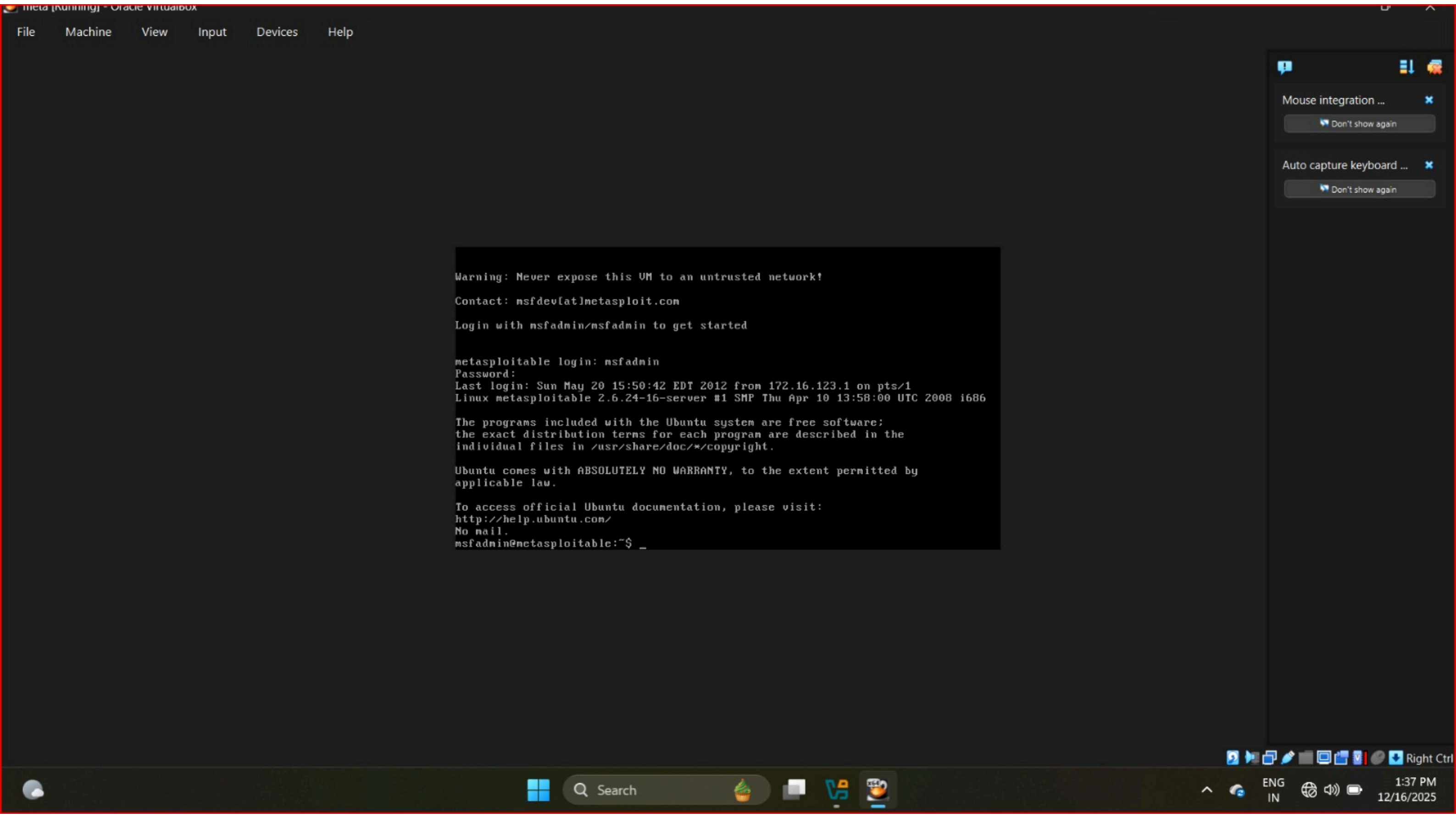
4:52 PM
12/15/2025


meta [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
metasploitable login: msfadmin
Password:
Last login: Sun May 20 15:50:42 EDT 2012 from 172.16.123.1 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ _
Right Ctrl
Search
ENG
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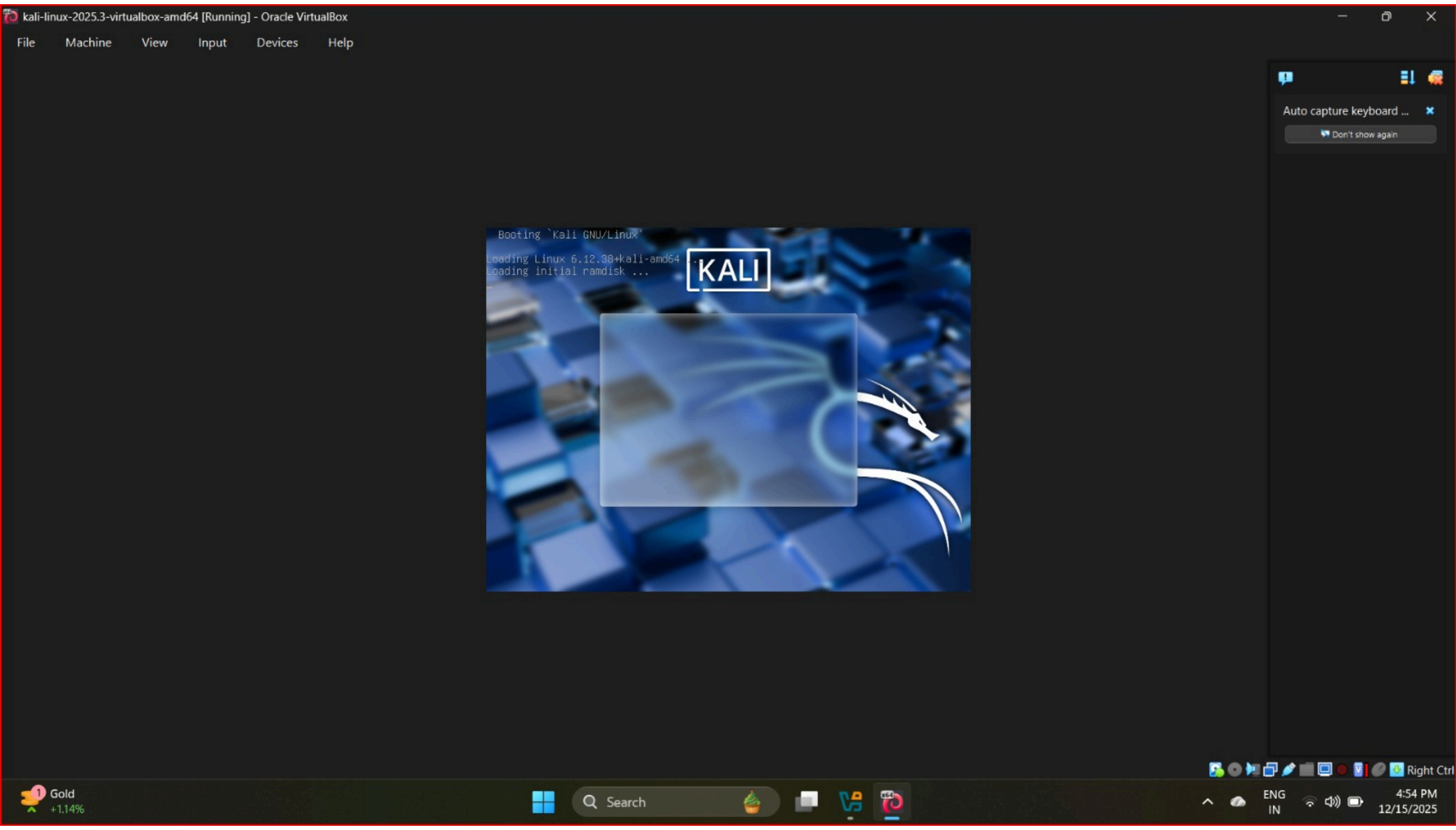
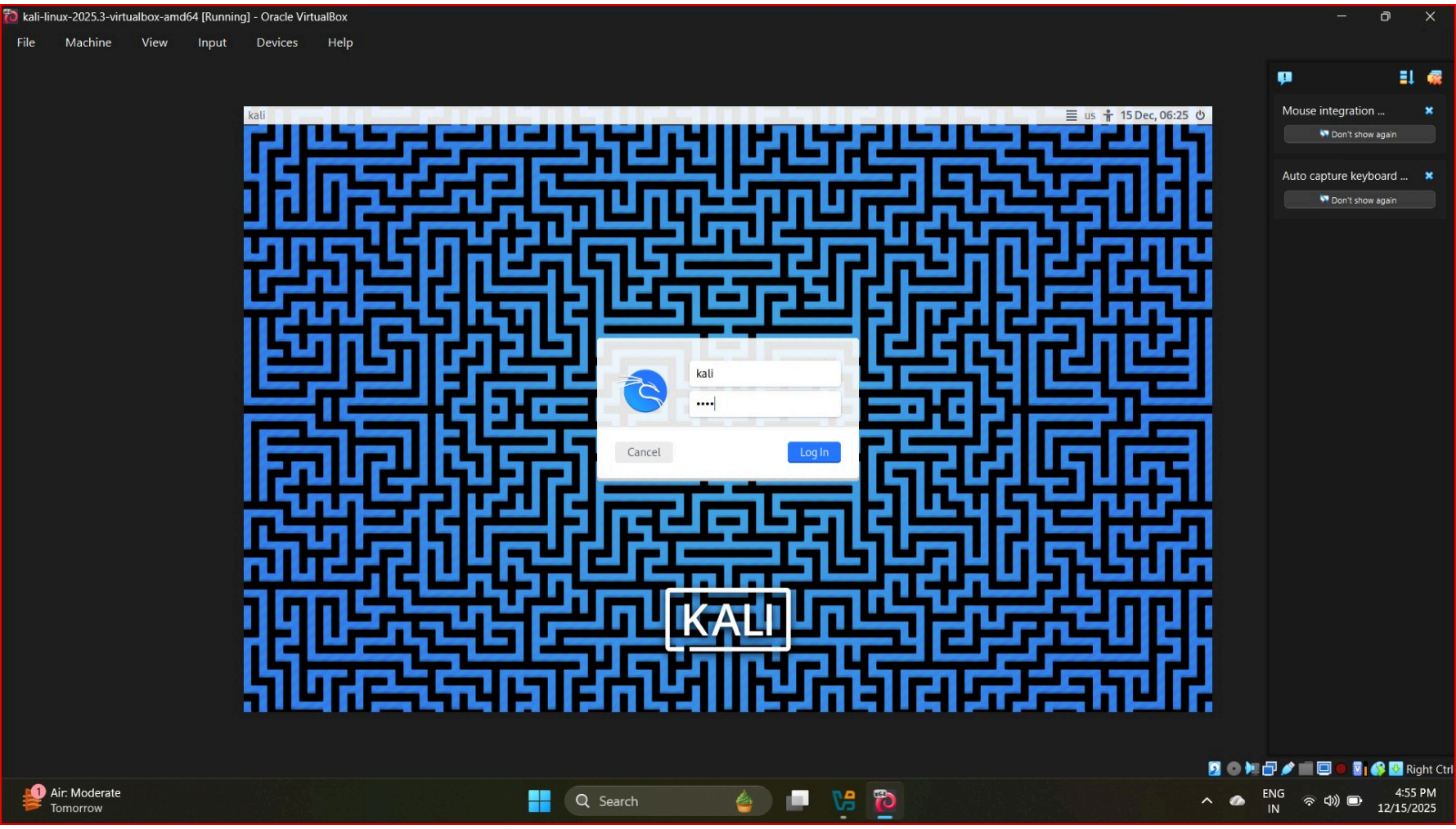
1:37 PM
12/16/2025


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Auto capture keyboard ...
Don't show again
Booting `Kali GNU/Linux'
Loading Linux 6.12.38+kali-amd64 ...
Loading initial ramdisk ...
KALI
Right Ctrl
Gold
+1.14%
Search
ENG
IN
4:54 PM
12/15/2025


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
kali
us
15 Dec, 06:25
kali
Cancel
Log In
KALI
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Air: Moderate
Tomorrow
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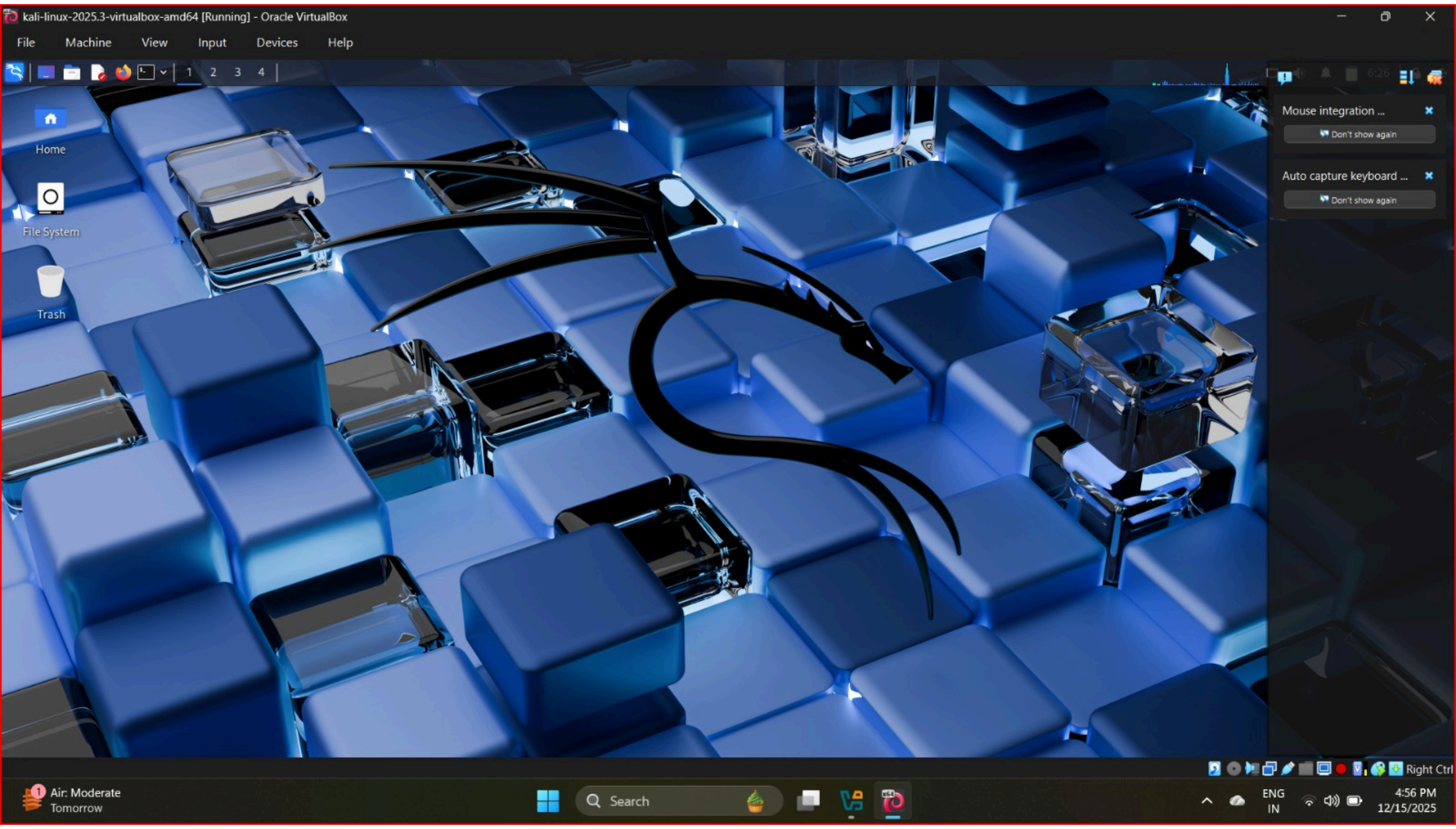
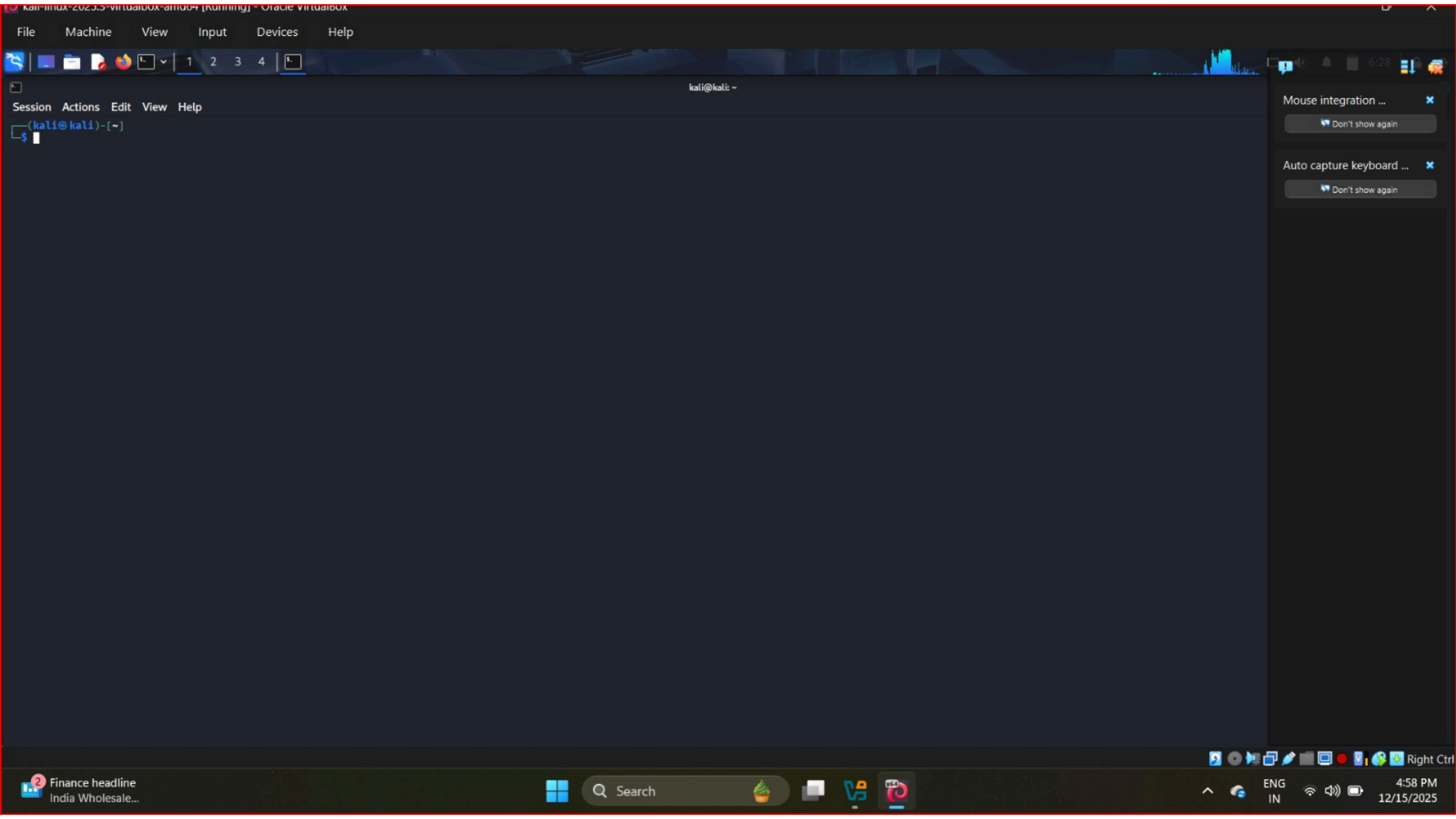
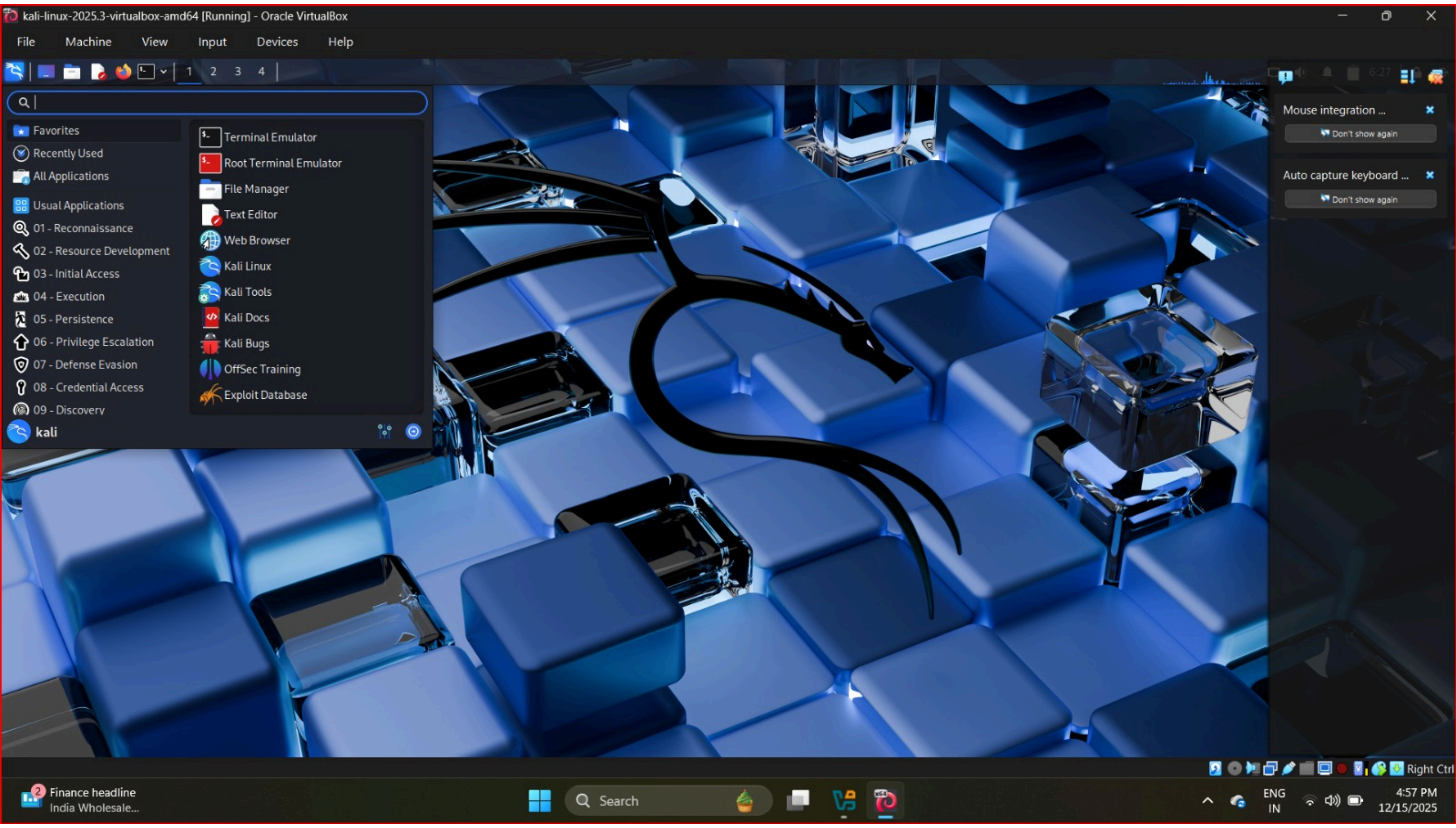
12/15/2025


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Home
File System
Trash
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Air: Moderate
Tomorrow
Search
ENG
IN
4:56 PM
12/15/2025


File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Finance headline
India Wholesale...
Search
ENG
IN
4:58 PM
12/15/2025


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Favorites
Recently Used
All Applications
Usual Applications
01 - Reconnaissance
02 - Resource Development
03 - Initial Access
04 - Execution
05 - Persistence
06 - Privilege Escalation
07 - Defense Evasion
08 - Credential Access
09 - Discovery
kali
Terminal Emulator
Root Terminal Emulator
File Manager
Text Editor
Web Browser
Kali Linux
Kali Tools
Kali Docs
Kali Bugs
OffSec Training
Exploit Database
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Finance headline
India Wholesale...
Search
ENG
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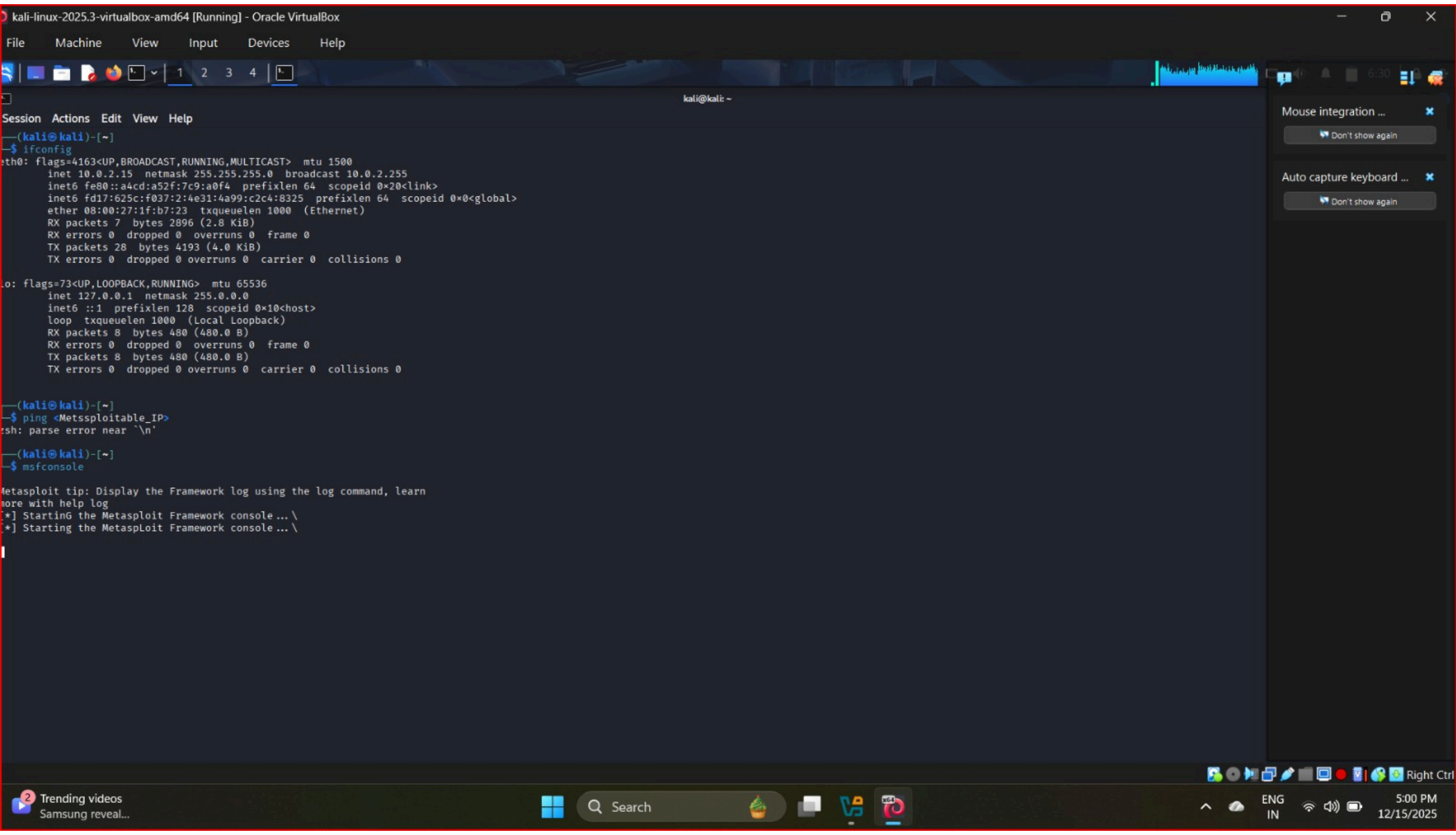
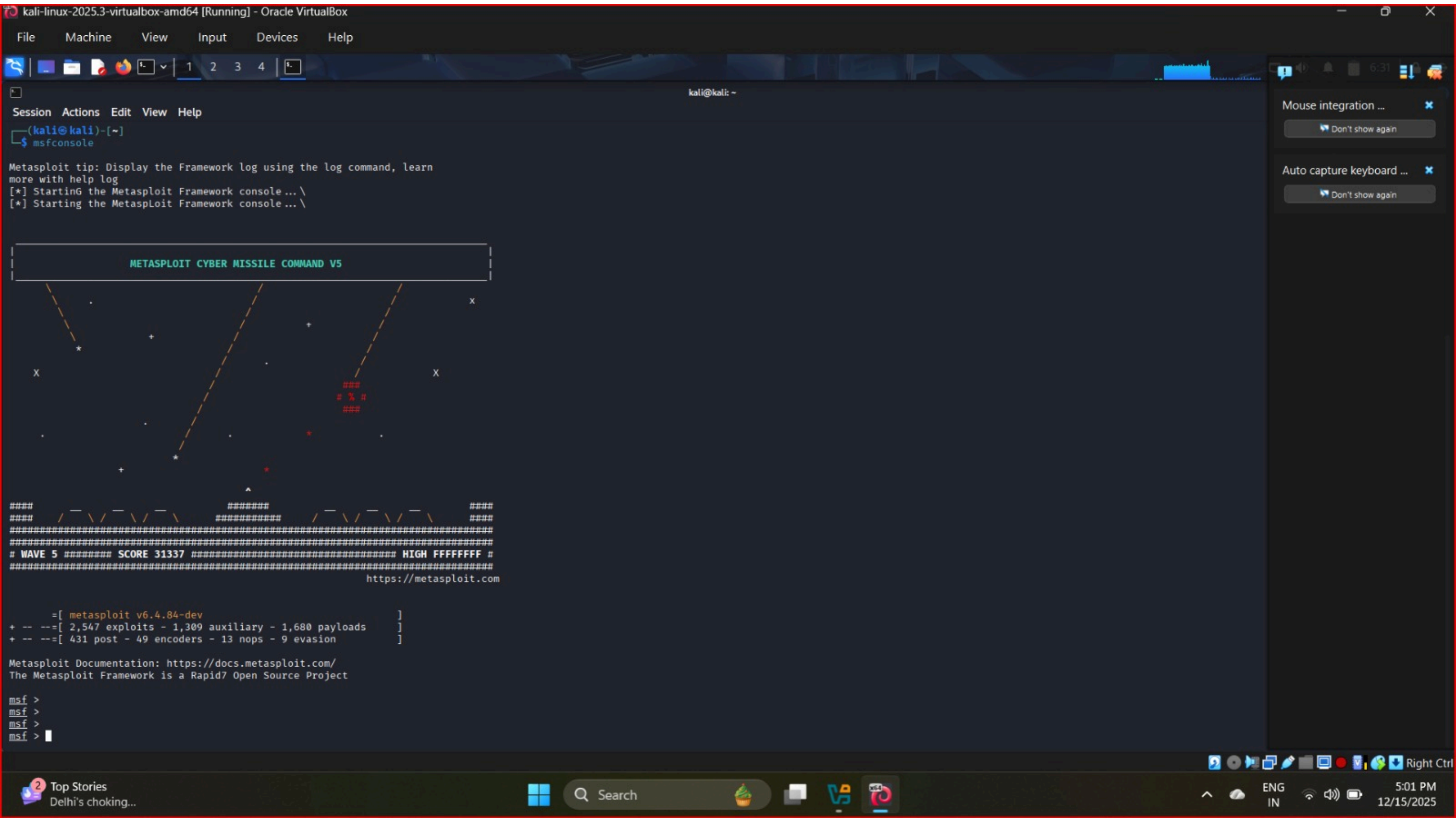
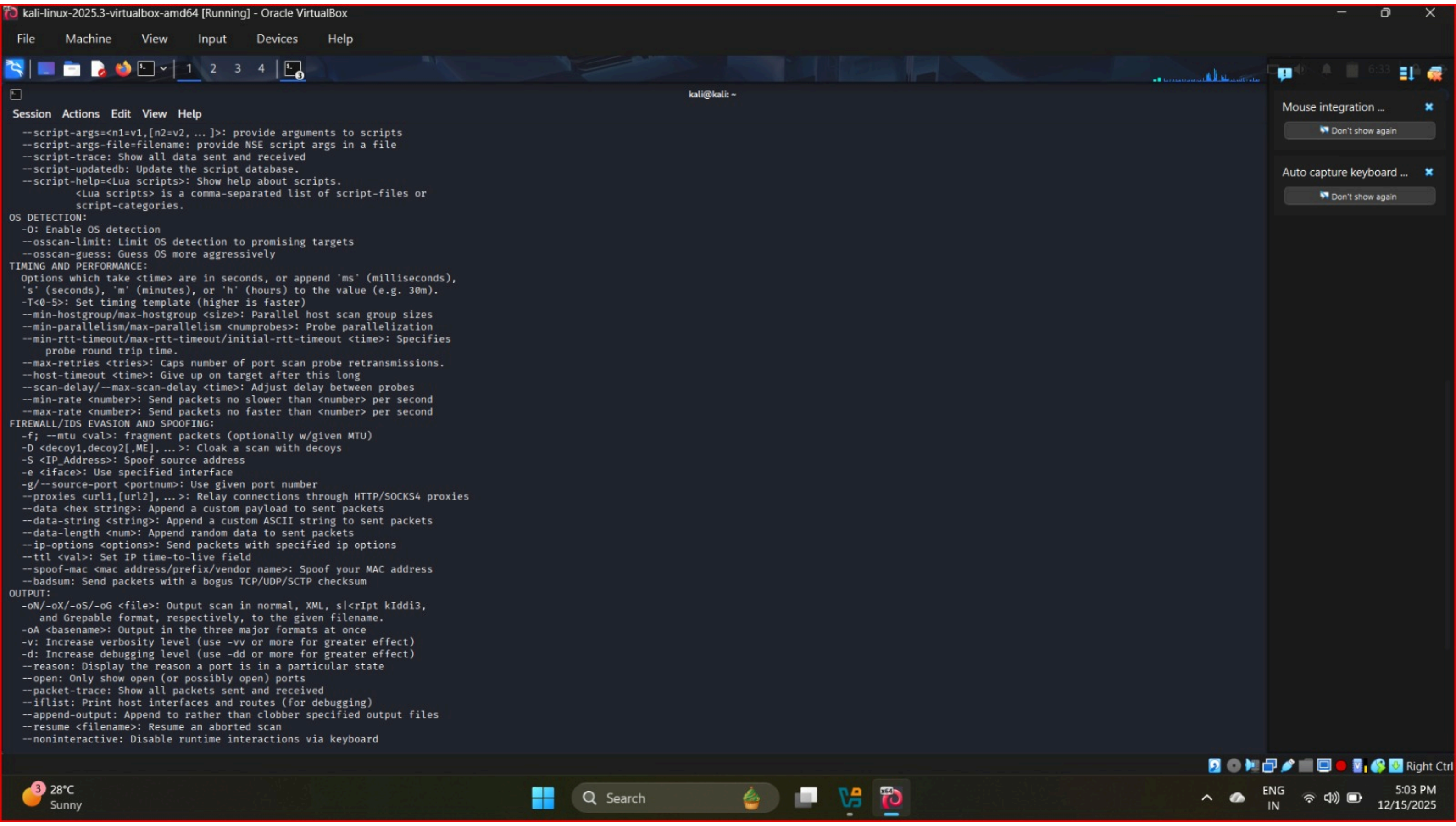
4:57 PM
12/15/2025


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fe80::a4cd:a52f:7c9:a0f4 prefixlen 64 scopeid 0x20<link>
inet6 fd17:625c:f037:2:4e31:4a99:c2c4:8325 prefixlen 64 scopeid 0x0<global>
ether 08:00:27:1f:b7:23 txqueuelen 1000 (Ethernet)
RX packets 7 bytes 2896 (2.8 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 28 bytes 4193 (4.0 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[~]
$ ping <Metssploitable_IP>
zsh: parse error near `\n'
(kali㉿kali)-[~]
$ msfconsole
Metasploit tip: Display the Framework log using the log command, learn
more with help log
[*] StartinG the Metasploit Framework console ... \
[*] Starting the MetaspLoit Framework console ... \
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Trending videos
Samsung reveal...
Search
ENG
IN
5:00 PM
12/15/2025



kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ msfconsole
Metasploit tip: Display the Framework log using the log command, learn
more with help log
[*] StartinG the Metasploit Framework console...\
[*] Starting the MetasploIt Framework console...\
METASPLOIT CYBER MISSILE COMMAND V5
WAVE 5 ######## SCORE 31337 ################################## HIGH FFFFFFFF
https://metasploit.com
=[metasploit v6.4.84-dev]
+ -- --=[2,547 exploits - 1,309 auxiliary - 1,680 payloads]
+ -- --=[431 post - 49 encoders - 13 nops - 9 evasion]
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project
msf >
msf >
msf >
msf >
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Top Stories
Delhi's choking...
Search
ENG
IN
5:01 PM
12/15/2025




```
  --script-args=<n1=v1,[n2=v2, ... ]>: provide arguments to scripts
  --script-args-file=filename: provide NSE script args in a file
  --script-trace: Show all data sent and received
  --script-updatedb: Update the script database.
  --script-help=<Lua scripts>: Show help about scripts.
           <Lua scripts> is a comma-separated list of script-files or
           script-categories.
OS DETECTION:
  -O: Enable OS detection
  --osscan-limit: Limit OS detection to promising targets
  --osscan-guess: Guess OS more aggressively
TIMING AND PERFORMANCE:
  Options which take <time> are in seconds, or append 'ms' (milliseconds),
  's' (seconds), 'm' (minutes), or 'h' (hours) to the value (e.g. 30m).
  -T<0-5>: Set timing template (higher is faster)
  --min-hostgroup/max-hostgroup <size>: Parallel host scan group sizes
  --min-parallelism/max-parallelism <numprobes>: Probe parallelization
  --min-rtt-timeout/max-rtt-timeout/initial-rtt-timeout <time>: Specifies
      probe round trip time.
  --max-retries <tries>: Caps number of port scan probe retransmissions.
  --host-timeout <time>: Give up on target after this long
  --scan-delay/--max-scan-delay <time>: Adjust delay between probes
  --min-rate <number>: Send packets no slower than <number> per second
  --max-rate <number>: Send packets no faster than <number> per second
FIREWALL/IDS EVASION AND SPOOFING:
  -f; --mtu <val>: fragment packets (optionally w/given MTU)
  -D <decoy1,decoy2[,ME], ... >: Cloak a scan with decoys
  -S <IP_Address>: Spoof source address
  -e <iface>: Use specified interface
  -g/--source-port <portnum>: Use given port number
  --proxies <url1,[url2], ... >: Relay connections through HTTP/SOCKS4 proxies
  --data <hex string>: Append a custom payload to sent packets
  --data-string <string>: Append a custom ASCII string to sent packets
  --data-length <num>: Append random data to sent packets
  --ip-options <options>: Send packets with specified ip options
  --ttl <val>: Set IP time-to-live field
  --spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address
  --badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
  -oN/-oX/-oS/-oG <file>: Output scan in normal, XML, s|<rIpt kIddi3,
     and Grepable format, respectively, to the given filename.
  -oA <basename>: Output in the three major formats at once
  -v: Increase verbosity level (use -vv or more for greater effect)
  -d: Increase debugging level (use -dd or more for greater effect)
  --reason: Display the reason a port is in a particular state
  --open: Only show open (or possibly open) ports
  --packet-trace: Show all packets sent and received
  --iflist: Print host interfaces and routes (for debugging)
  --append-output: Append to rather than clobber specified output files
  --resume <filename>: Resume an aborted scan
  --noninteractive: Disable runtime interactions via keyboard
```

kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

kali@kali: ~

Session Actions Edit View Help

```
$ nmap -h
Nmap 7.95 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script=default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
           directories, script-files or script-categories
  --script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts
  --script-args-file=filename: provide NSE script args in a file
  --script-trace: Show all data sent and received
```

Mouse integration ...
Don't show again

Auto capture keyboard ...
Don't show again

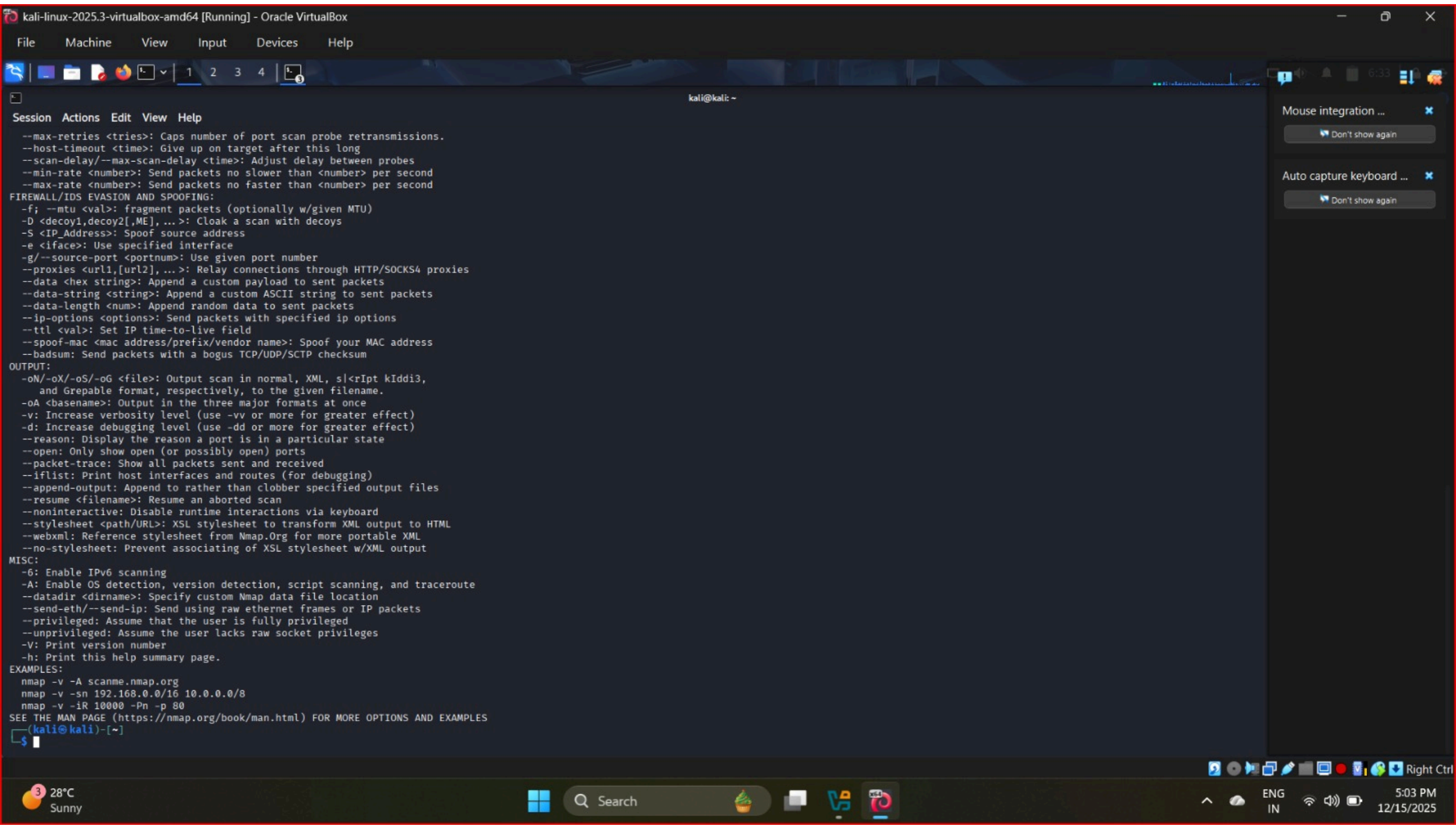
Right Ctrl

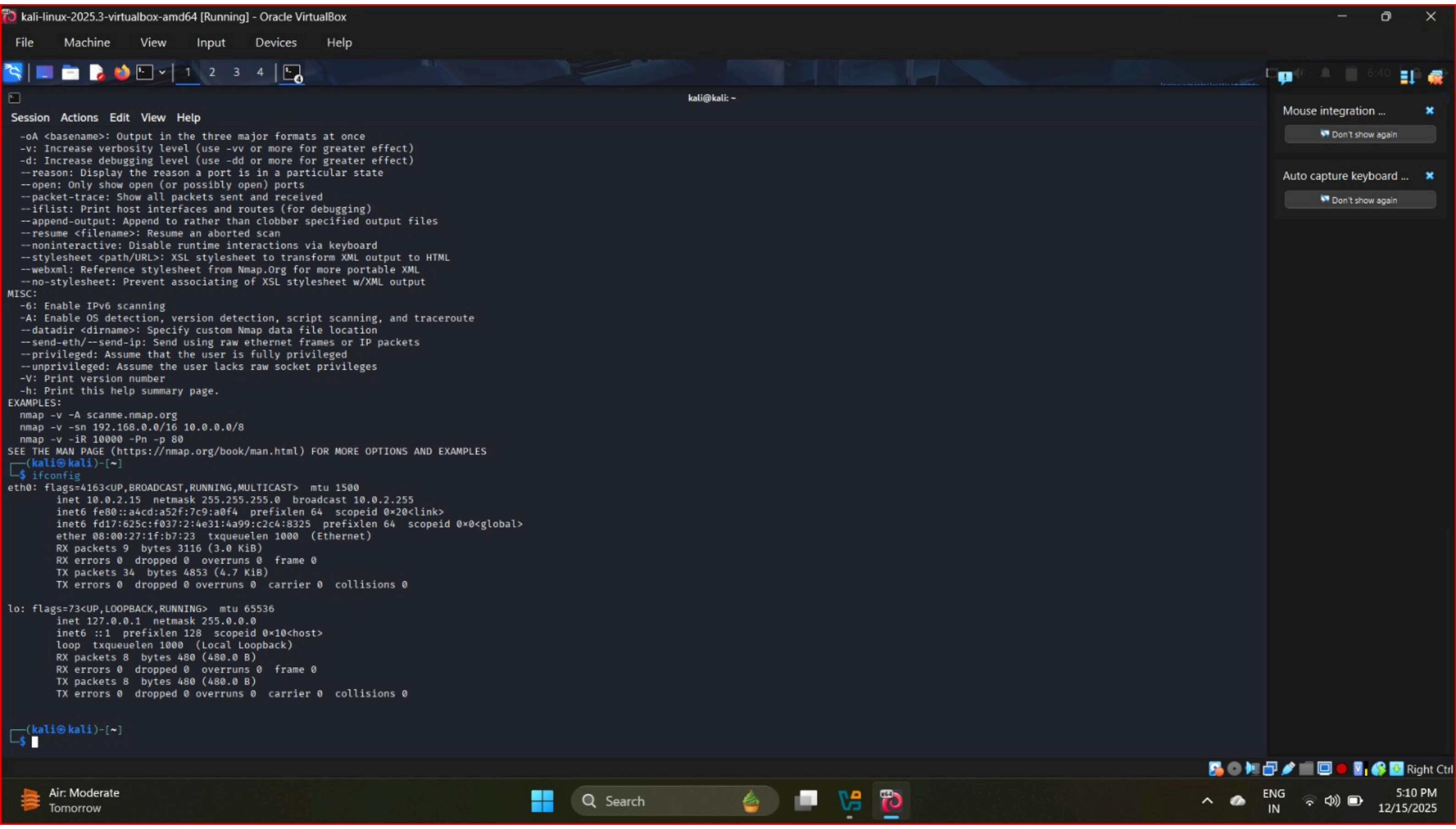
28°C
Sunny

Search

ENG
IN

5:02 PM
12/15/2025


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
--max-retries <tries>: Caps number of port scan probe retransmissions.
--host-timeout <time>: Give up on target after this long
--scan-delay/--max-scan-delay <time>: Adjust delay between probes
--min-rate <number>: Send packets no slower than <number> per second
--max-rate <number>: Send packets no faster than <number> per second
FIREWALL/IDS EVASION AND SPOOFING:
-f; --mtu <val>: fragment packets (optionally w/given MTU)
-D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys
-S <IP_Address>: Spoof source address
-e <iface>: Use specified interface
-g/--source-port <portnum>: Use given port number
--proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies
--data <hex string>: Append a custom payload to sent packets
--data-string <string>: Append a custom ASCII string to sent packets
--data-length <num>: Append random data to sent packets
--ip-options <options>: Send packets with specified ip options
--ttl <val>: Set IP time-to-live field
--spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, s|<rIpt kIddi3,
and Grepable format, respectively, to the given filename.
-oA <basename>: Output in the three major formats at once
-v: Increase verbosity level (use -vv or more for greater effect)
-d: Increase debugging level (use -dd or more for greater effect)
--reason: Display the reason a port is in a particular state
--open: Only show open (or possibly open) ports
--packet-trace: Show all packets sent and received
--iflist: Print host interfaces and routes (for debugging)
--append-output: Append to rather than clobber specified output files
--resume <filename>: Resume an aborted scan
--noninteractive: Disable runtime interactions via keyboard
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
--webxml: Reference stylesheet from Nmap.Org for more portable XML
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
-6: Enable IPv6 scanning
-A: Enable OS detection, version detection, script scanning, and traceroute
--datadir <dirname>: Specify custom Nmap data file location
--send-eth/--send-ip: Send using raw ethernet frames or IP packets
--privileged: Assume that the user is fully privileged
--unprivileged: Assume the user lacks raw socket privileges
-V: Print version number
-h: Print this help summary page.
EXAMPLES:
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
(kali㉿kali)-[~]
$
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
28°C
Sunny
Search
ENG
IN
5:03 PM
12/15/2025



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SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fe80::a4cd:a52f:7c9:a0f4 prefixlen 64 scopeid 0x20<link>
inet6 fd17:625c:f037:2:4e31:4a99:c2c4:8325 prefixlen 64 scopeid 0x0<global>
ether 08:00:27:1f:b7:23 txqueuelen 1000 (Ethernet)
RX packets 9 bytes 3116 (3.0 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 34 bytes 4853 (4.7 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[~]
$
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl
Air: Moderate
Tomorrow
Search
ENG
IN
5:10 PM
12/15/2025